

Oliver Dippel

Research Scientist — Reinforcement Learning & Foundation Models

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Technical Profile

Ph.D. researcher specializing in In-Context Reinforcement Learning (ICRL) and decision-making foundation models. Proven track record in designing principled RL architectures and Transformer-based policies using JAX, Ray and PyTorch. Expertise in bridging sequential decision-making with LLM-driven reasoning, focusing on alignment (RLHF), task generalization, and scalable distributed training. *Note: Core Ph.D. and industry codebases are proprietary due to corporate funding; technical details are available via peer-reviewed publications.*

Research & Engineering Experience

Doctoral Researcher — RL & Foundation Models

Dec 2021 – Present

University of Liverpool / CDT in Distributed Algorithms

Liverpool, UK

- Architected the Heuristic Transformer (HT), a novel Bayesian in-context RL framework utilizing latent stochastic variables for zero-shot task adaptation.
- Engineered large-scale, reproducible training pipelines in JAX and PyTorch, supporting high-throughput experimentation on HPC clusters.
- Investigated posterior regularization and reward design with direct applications to RLHF and agent alignment.
- Developed uncertainty-aware policy priors that enable rapid adaptation to non-stationary environments without gradient updates.
- Published in top-tier venues, e.g. AAMAS.

Research Engineer — LLM Agents & Generative Systems

Sep 2023 – Dec 2024

ZeroAI - Part-Time, alongside PhD

Liverpool, UK

- Developed LLM-based agents for structured reasoning and interpretable decision-making in safety-critical autonomous driving environments.
- Integrated Diffusion Models for trajectory forecasting, improving model-based decision support in high-dimensional state spaces.
- Optimized closed-loop simulation frameworks to evaluate agent robustness and failure modes under diverse edge cases.

Statistical Consulting & Teaching

Jan 2021 – Present

Freelancer / statistiknachhilfe.ch

Zurich, CH

- Advise Master's and Bachelor students on experimental design, Bayesian inference, and statistical rigor.
- Curate and teach advanced modules on probabilistic machine learning, focusing on experimental design and robust implementation in R and Python.

Data Science Intern — Applied ML Systems

Jun 2020 – Dec 2020

Novartis Pharma AG

Basel, CH

- Built production-grade data pipelines for ML-driven clinical decision support within highly regulated environments.

Key Research Contribution

Heuristic Transformer (ICRL Framework)

Developed a belief-augmented Transformer architecture that addresses the reward ambiguity problem in zero-shot settings by utilising a stochastic latent variable to represent reward distributions, enabling inference of task-specific heuristics entirely in-context.

Technical Skills

Frameworks: JAX (Haiku/Flax), PyTorch, Ray, Weights & Biases, Docker, Linux/HPC.

Specializations: Reinforcement Learning (Deep, Bayesian, Multi-Agent), Foundation Models (Transformers, ICL), Generative Modeling (Diffusion, VAE), RLHF, Policy Optimization, Distributed Training.

Education

Ph.D. Electrical & Control Engineering — University of Liverpool 2021 – 2025

Thesis: Bayesian RL for High-Dimensional Continuous Control (Industry Funded)

M.Sc. Applied Statistics — University of Göttingen 2017 – 2020

Thesis: Deep Unsupervised Data Compression for Large-Scale Financial Datasets

B.A. Economics — University of Göttingen 2014 – 2017

Selected Publications

- *Heuristic Transformer: Belief-Augmented In-Context Reinforcement Learning* AAMAS, 2026
- *Contextual Transformers for Goal-Oriented Reinforcement Learning* SGAI, 2024
- *Deep Reinforcement Learning for Continuous Control of Material Thickness* SGAI, 2023

Languages

German (Native), **English** (Full Professional), **French** (Intermediate)